

Fig. 4: Power supply circuit

is interfaced to the electrical appliance (lamp, fan, motor) with mechanical switch K2.

Thus the electrical appliance can be

operated either electronically by applying logic high or low to the control signal, or by the mechanical switch, which can be mounted on a wall traditionally.

Resistor R7
in series with
capacitor C1

across TRIAC1, and resistor R8 in series with capacitor C2 across TRIAC2 form snubber circuits, which ensure that switching of the inductive load

does not create peak voltages across the TRIAC terminals. Only one gate (N1) of IC1 is used here. The remaining inverters of IC1 can be used for other solid-state two-way switches not shown in the circuit.

The +5V DC power supply for this circuit can be derived from 230V AC mains using the circuit shown in Fig. 4.

Assemble both the circuits (Figs 3 and 4) on a single general-purpose PCB and enclose in a suitable case. Connect mains power cord for input. Use a proper socket for the appliance and switch to enable and disable the circuit. Connect mechanical switch K2 through external wires and install the unit at a suitable place. ●

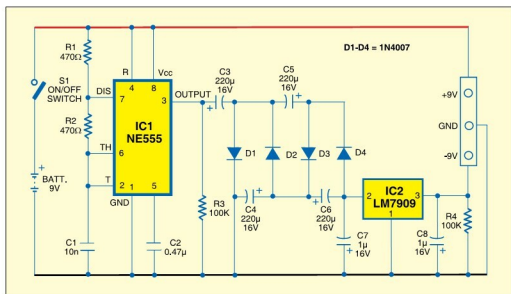
DUAL SUPPLY FROM SINGLE BATTERY SOURCE

■ **DILEEP CHACKO**

Dual-voltage power supply is required especially for powering the op-amps and some of the instrumentation amplifiers. A few low-power audio pre-amplifiers also use dual-voltage supply.

This circuit provides both +9V as well as -9V supply from a single 9V battery. It consists of timer IC NE555 (IC1), negative-voltage regulator IC 7909 (IC2) and a few discrete components.

Timer IC NE555 is configured as a free-running oscillator. It produces around 100kHz frequency. An au-



dio system would not pick up this frequency as it is above the audible range. Therefore a Villard cascade voltage multiplier is used as the voltage booster. The boosted voltage is made ripple-free and regulated to -9V by negative-voltage regulator LM7909. The frequency of the squarewave gen-

erator depends on the values of C1, R1 and R2. The frequency of the signal may slightly change with the temperature variation but the change in frequency does not affect the output voltage of the circuit.

Capacitors C3, C4, C5 and C6 and diodes D1, D2, D3 and D4 form the Villard cascade voltage multiplier. The output voltage from the Villard cascade voltage multiplier is fed to linear negative-voltage regulator IC2 to elimi-

